

PROGRAM 27

AIM

To implement a simple Medical Diagnosis Expert System in Prolog that identifies a possible disease based on the symptoms provided by the user.

ALGORITHM

1. Start the program.
2. Define diseases and their corresponding symptoms as facts.
3. Define rules to identify diseases based on symptoms.
4. Accept the symptoms from the user.
5. Compare the given symptoms with the knowledge base.
6. Identify the matching disease.
7. Display the possible diagnosis.
8. Stop the program.

PROGRAM

```
symptom(fever, flu).  
symptom(cough, flu).  
symptom(cold, flu).  
symptom(headache, flu).  
symptom(sneezing, cold).  
symptom(runny_nose, cold).  
symptom(cough, cold).  
symptom(thirst, diabetes).  
symptom(frequent_urination, diabetes).  
symptom(fatigue, diabetes).  
diagnose(Disease, Symptoms) :-  
    findall(S, symptom(S, Disease), Symptoms).
```

OUTPUT

```
47 ?- consult(['c:/Users/Hema Neeпа/OneDrive/Documents/Medical_Diagnosis.pl']).  
true.
```

```
48 ?- diagnose(flu, Symptoms).  
Symptoms = [fever, cough, cold, headache].
```

```
49 ?- diagnose(cold, Symptoms).  
Symptoms = [sneezing, runny_nose, cough].
```

```
50 ?- diagnose(diabetes, Symptoms).  
Symptoms = [thirst, frequent_urination, fatigue].
```

RESULT

The Medical Diagnosis Expert System was successfully implemented in Prolog, and possible diseases were identified based on their associated symptoms